


```

1501A C7 45 08 6C 12 ED 3C      mov     [ebp+enc+28h], 3CED126Ch
15021 C7 45 DC 31 05 E0 F5      mov     [ebp+enc+2Ch], 0F5E00531h
15028 C7 45 A4 00 00 00 00      mov     [ebp+FlagInput], 0
1502F 6A 64                      push    64h ; 'd' ; Size
15031 6A 00                      push    0 ; Val
15033 8D 85 38 FF FF FF          lea     eax, [ebp+Input] ; void *
15039 50                          push    eax
1503A E8 CE C1 FF FF              call    j__memset
1503A
1503F 83 C4 0C                    add     esp, 0Ch
15042 68 74 DC 41 00            push    offset _Val ; "Please input your flag:"
15047 A1 FC 10 42 00            mov     eax, ds:___imp_?cout@std@@3V?$basic_ostream@DU?$char_traits@D@std@@@1@A ; std::ostream std::cout
1504C 50                          push    eax ; _Ostr
1504D E8 E8 C1 FF FF              call    j_??$?6U?$char_traits@D@std@@@std@@YAAAV?$basic_ostream@DU?$char_traits@D@std@@@0@AAV10@PBD@Z ; std::operator<<<std::char,
1504D
15052 83 C4 08                    add     esp, 8
15055 8D 85 38 FF FF FF          lea     eax, [ebp+Input]
15058 50                          push    eax ; _Str
1505C 8B 0D 98 10 42 00          mov     ecx, ds:___imp_?cin@std@@3V?$basic_istream@DU?$char_traits@D@std@@@1@A ; std::istream std::cin
15062 51                          push    ecx ; _Istr
15063 E8 60 C0 FF FF              call    j_??$?5DU?$char_traits@D@std@@@std@@YAAAV?$basic_istream@DU?$char_traits@D@std@@@0@AAV10@PAD@Z ; std::operator>>>char,std
15063
15068 83 C4 08                    add     esp, 8
15068 90                          nop
1506C 90                          nop
1506D 90                          nop
1506E 90                          nop
1506F 90                          nop
15070 90                          nop
15071 90                          nop
15072 90                          nop
15073 90                          nop
15074 90                          nop
15075
15075 f1_0:
15075 90                          nop
15075
15075 ; -----
15076 90                          db  90h
15077 90                          db  90h
15078 90                          db  90h
15079 90                          db  90h
1507A 90                          db  90h
1507B
1507B ; -----
1507B 8D 85 38 FF FF FF          lea     eax, [ebp+Input]
15081 89 45 A4                    mov     [ebp+FlagInput], eax
15084 C7 85 2C FF FF FF 00 00 00 00 mov     [ebp+i], 0
1508E EB 0F                      jmp     short loc_41509F
1508E
15090 ; -----

```

去花

```

2 {
3   int i; // [esp+D0h] [ebp-D4h]
4   char _Str[108]; // [esp+DCh] [ebp-C8h] BYREF
5   unsigned int *Input; // [esp+148h] [ebp-5Ch]
6   unsigned int enc[14]; // [esp+154h] [ebp-50h] BYREF
7   unsigned int Key[5]; // [esp+18Ch] [ebp-18h] BYREF
8
9   Key[0] = 287454020;
10  Key[1] = 1432778632;
11  Key[2] = -1716864052;
12  Key[3] = -571539695;
13  enc[0] = 849219247;
14  enc[1] = 1782154081;
15  enc[2] = 839590394;
16  enc[3] = 570472433;
17  enc[4] = 772302708;
18  enc[5] = 1961553718;
19  enc[6] = -386987485;
20  enc[7] = -1345016732;
21  enc[8] = -1829160445;
22  enc[9] = -1283810305;
23  enc[10] = 1022169708;
24  enc[11] = -169867983;
25  Input = 0;
26  j__memset(_Str, 0, 0x64u);
27  std::operator<<<std::char_traits<char>>(std::cout, "Please input your flag:");
28  std::operator>>>char,std::char_traits<char>>(std::cin, _Str);
29  Input = (unsigned int *)_Str;
30  for ( i = 0; i < 6; ++i )
31  {
32      Tea(&Input[2 * i], Key);
33      if ( Check(enc, Input) )
34          std::operator<<<std::char_traits<char>>(std::cout, &byte_41DC90);
35      else
36          std::operator<<<std::char_traits<char>>(std::cout, &byte_41DCA0);
37  }
38  return 0;
39 }

```

然后tea里面还有花同理

```

.text:004148D9 53          push     ebx
.text:004148DA 56          push     esi
.text:004148DB 57          push     edi
.text:004148DC
.text:004148DC          __EncStackInitStart_28:
.text:004148DC 8D 7D D0    lea      edi, [ebp+var_30]
.text:004148DF B9 0C 00 00 00 mov     ecx, 0Ch
.text:004148E4 B8 CC CC CC CC mov     eax, 0CCCCCCCCh
.text:004148E9 F3 AB      rep stosl
.text:004148EB
.text:004148EB          __EncStackInitEnd_28:
.text:004148EB C7 45 F8 00 00 00 00 mov     [ebp+sum], 0
.text:004148F2 B8 04 00 00 00 mov     eax, 4
.text:004148F7 68 C8 00    imul     ecx, eax, 0
.text:004148FA 88 55 08    mov     edx, [ebp+flag]
.text:004148FD 8B 04 0A    mov     eax, [edx+ecx]
.text:00414900 89 45 E0    mov     [ebp+v1], eax
.text:00414903 B8 04 00 00 00 mov     eax, 4
.text:00414908 C1 E0 00    shl     eax, 0
.text:0041490B 8B 4D 08    mov     ecx, [ebp+flag]
.text:0041490E 8B 14 01    mov     edx, [ecx+eax]
.text:00414911 89 55 D4    mov     [ebp+v2], edx
.text:00414914 74 03      jz       short near ptr loc_414918+1
.text:00414914
.text:00414914          jnz     short near ptr loc_414918+1
.text:00414916 75 01
.text:00414916
.text:00414918
loc_414918:                                ; CODE XREF: Tea(uint *,uint const *)+44tj
.text:00414918                                ; Tea(uint *,uint const *)+46tj
.text:00414918 89 E8      mov     eax, ebp
.text:00414918
; -----
.text:0041491A 00 00      dw 0
.text:0041491C 00 00      db 2 dup(0)
; -----
.text:0041491E
f1:
.text:0041491E 80 04 24 06 add     byte ptr [esp+], 6
.text:00414922 C3          retn
.text:00414922
; -----
.text:00414923 89          db 89h
; -----
.text:00414924
; -----
.text:00414924 C7 45 EC 00 00 00 00 mov     [ebp+i], 0
.text:0041492B EB 09      jmp     short loc_414936
; -----
loc_41492D:                                ; CODE XREF: Tea(uint *,uint const *)+E6tj
.text:0041492D 8B 45 EC    mov     eax, [ebp+i]
.text:00414930 83 C0 01    add     eax, 1
.text:00414933 89 45 EC    mov     [ebp+i], eax
; -----
loc_414936:                                ; CODE XREF: Tea(uint *,uint const *)+5Btj
.text:00414936 cmp     [ebp+i], 20h ; ' '
.text:00414936 7D 7E      jnz     short loc_41493B

```

IDA 56	push esi
IDB 57	push edi
IDC	
IDC	__\$EncStackInitStart_28:
IDC 8D 7D D0	lea edi, [ebp+var_30]
IDF B9 0C 00 00 00	mov ecx, 0Ch
IE4 B8 CC CC CC CC	mov eax, 0CCCCCCCCh
IE9 F3 AB	rep stosd
IEB	
IEB	__\$EncStackInitEnd_28:
IEB C7 45 F8 00 00 00 00	mov [ebp+sum], 0
IF2 B8 04 00 00 00	mov eax, 4
IF7 6B C8 00	imul ecx, eax, 0
IFA 8B 55 08	mov edx, [ebp+flag]
IFD 8B 04 0A	mov eax, [edx+ecx]
I00 89 45 E0	mov [ebp+v1], eax
I03 B8 04 00 00 00	mov eax, 4
I08 C1 E0 00	shl eax, 0
I0B 8B 4D 08	mov ecx, [ebp+flag]
I0E 8B 14 01	mov edx, [ecx+eax]
I11 89 55 D4	mov [ebp+v2], edx
I14 74 03	jz short near ptr loc_414918+1
I14	
I16 75 01	jnz short near ptr loc_414918+1
I16	
I18	
I18	loc_414918: ; CODE XREF: Tea(uint *,uint const *)+44↑j
I18	; Tea(uint *,uint const *)+46↑j
I18 89 E8	mov eax, ebp
I18	
I18	; -----
IA 00 00	dw 0
IC 00 00	db 2 dup(0)
IE	; -----
IE	
IE	f1:
IE 80 04 24 06	add byte ptr [esp+0], 6
I22 C3	retn
I22	
I22	; -----
I23 89	db 89h
I24	; -----
I24 C7 45 EC 00 00 00 00	mov [ebp+i], 0
I2B EB 09	jmp short loc_414936
I2B	
I2D	; -----
I2D	
I2D	loc_41492D: ; CODE XREF: Tea(uint *,uint const *)+E6↓j
I2D 8B 45 EC	mov eax, [ebp+i]
I30 83 C0 01	add eax, 1
I33 89 45 EC	mov [ebp+i], eax
I33	
I36	
I36	loc_414936: ; CODE XREF: Tea(uint *,uint const *)+E8↑j

nop完

function	IA View-A	Pseudocode-A	Hex View-1	Structures
t:004148D0		sum= dword ptr -8		
t:004148D0		flag= dword ptr 8		
t:004148D0		Key= dword ptr 0Ch		
t:004148D0				
t:004148D0 55		push ebp		
t:004148D1 88 EC		mov ebp, esp		
t:004148D3 81 EC F0 00 00 00		sub esp, 0F0h		
t:004148D9 53		push ebx		
t:004148DA 56		push esi		
t:004148DB 57		push edi		
t:004148DC				
t:004148DC		__\$EncStackInitStart_28:		
t:004148DC 8D 7D 00		lea edi, [ebp+var_30]		
t:004148DF B9 0C 00 00 00		mov ecx, 0Ch		
t:004148E4 B8 CC CC CC CC		mov eax, 0CCCCCCCCh		
t:004148E9 F3 AB		rep stosd		
t:004148E8				
t:004148E8		__\$EncStackInitEnd_28:		
t:004148E8 C7 45 F8 00 00 00 00		mov [ebp+sum], 0		
t:004148F2 B8 04 00 00 00		mov eax, 4		
t:004148F7 6B C8 00		imul ecx, eax, 0		
t:004148FA 8B 55 08		mov edx, [ebp+flag]		
t:004148FD 8B 04 0A		mov eax, [edx+ecx]		
t:00414900 89 45 E0		mov [ebp+v1], eax		
t:00414903 B8 04 00 00 00		mov eax, 4		
t:00414908 C1 E0 00		shl eax, 0		
t:0041490B 8B 4D 08		mov ecx, [ebp+flag]		
t:0041490E 8B 14 01		mov edx, [ecx+eax]		
t:00414911 89 55 D4		mov [ebp+v2], edx		
t:00414914 90		nop		
t:00414915 90		nop		
t:00414916 90		nop		
t:00414917 90		nop		
t:00414918 90		nop		
t:00414919 90		nop		
t:0041491A 90		nop		
t:0041491B 90		nop		
t:0041491C 90		nop		
t:0041491D 90		nop		
t:0041491E				
t:0041491E		f1:		
t:0041491E 90		nop		
t:0041491F 90		nop		
t:00414920 90		nop		
t:00414921 90		nop		
t:00414922 90		nop		
t:00414923 90		nop		
t:00414924 C7 45 EC 00 00 00 00		mov [ebp+i], 0		
t:00414928 EB 09		jmp short loc_414936		
t:00414928				
t:0041492D				
t:0041492D				
t:0041492D		loc_41492D:		
t:0041492D 8B 45 EC		mov eax, [ebp+i]		
t:00414930 83 C0 01		add eax, 1		
t:00414933 89 45 EC		mov [ebp+i], eax		
t:00414933				
t:00414936				
t:00414936		loc_414936:		
t:00414936 83 7D EC 20		cmp [ebp+i], 20h ; ' '		
t:0041493A 7D 7F		jge short loc_41498B		

IDA View-A	Pseudocode-A	Hex View-1
1 void __cdecl Tea(unsigned int *flag, const unsigned int *Key)		
2 {		
3 unsigned int v2; // [esp+D0h] [ebp-2Ch]		
4 unsigned int v1; // [esp+DCh] [ebp-20h]		
5 int i; // [esp+E8h] [ebp-14h]		
6 unsigned int sum; // [esp+F4h] [ebp-8h]		
7		
8 sum = 0;		
9 v1 = *flag;		
10 v2 = flag[1];		
11 for (i = 0; i < 32; ++i)		
12 {		
13 sum += 1131796;		
14 v1 += (Key[1] + (v2 >> 5)) ^ (sum + v2) ^ (*Key + 16 * v2);		
15 v2 += (Key[3] + (v1 >> 5)) ^ (sum + v1) ^ (Key[2] + 16 * v1);		
16 }		
17 *flag = v1;		
18 flag[1] = v2;		
19 }		

去除发现是tea

```
#define _CRT_SECURE_NO_WARNINGS 1
#include <stdio.h>
```

```

#include <stdlib.h>
#include <stdint.h>
#include <string.h>
#define DELTA 0x114514

void DTea(uint32_t* flag, const uint32_t* Key) {
    uint32_t sum = DELTA * 32;
    int i;
    uint32_t v1 = flag[0];
    uint32_t v2 = flag[1];
    for (i = 0; i < 32; i++) {
        v2 -= ((v1 << 4) + Key[2]) ^ (v1 + sum) ^ ((v1 >> 5) + Key[3]);
        v1 -= ((v2 << 4) + Key[0]) ^ (v2 + sum) ^ ((v2 >> 5) + Key[1]);
        sum -= DELTA;
    }
    flag[0] = v1;
    flag[1] = v2;
}

int main() {
    uint32_t Key[4] = {0x11223344, 0x55667788, 0x99AABBCC, 0xDDEEFF11 };
    uint32_t enc[50] =
{0x329e0eaf,0x6a398361,0x320b21fa,0x2200b7f1,0x2e086774,0x74eaeef36,0xe8ef0a23,0xafd4ac64,0x92f93a03,0xb37a9bff,0x3ced126c,0xf5e00531};

    for (int i = 0; i < 6; i++) {
        DTea(&enc[i * 2], Key);
    }
    puts((char *)enc);

    return 0;
}

```